



XLC-WYLDFYRE



OUR TEAM:

Founded in 2009, XLC has been dedicated to advancing the field of robotics through innovation and teamwork. Since 2021, Our team has continued this mission with our talented team members, each bringing unique skills and expertise to our collective effort.

ABSTRACT:

We began designing the first generation of our robot in late 2022 to compete in our home country's 2023 RoboCup Junior competition. Since then, we have developed many new skills, tackled design challenges, and learned from our failures. Last year's competition turned out great for us, as we achieved 3rd place despite our robot's flawed designs, numerous malfunctions, and our initial struggles with coding. This year, we have worked extremely hard and developed a new design for our top PCBs. We switched microcontrollers, added new sensors for IR ball detection and a camera, upgraded the gyro sensor, and integrated a kicker and dribbler.



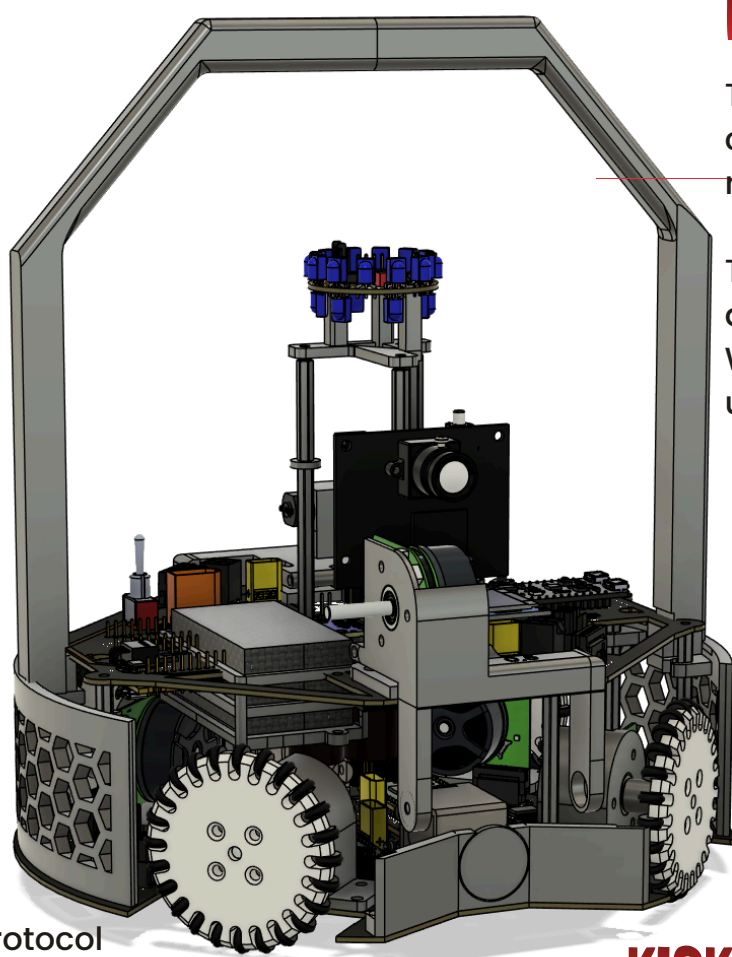
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MECHANICAL DESIGN:

First, we began designing our robot as a 3D model in Fusion 360. This proved to be an excellent tool because it allowed us to create the robot according to the specified dimensions outlined in the competition rules. We decided that the main structure of our robot would be made of PCBs.

HOMEMADE OMNIWHEELS:

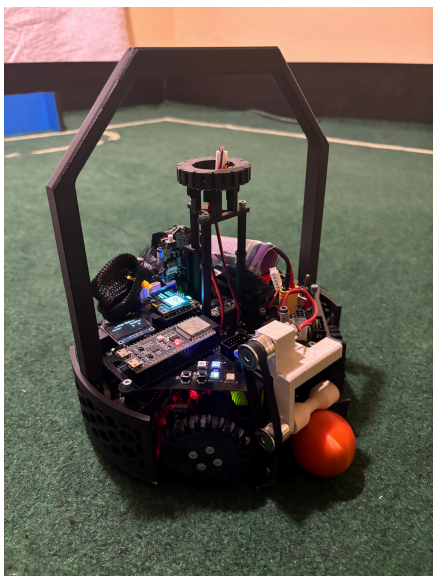
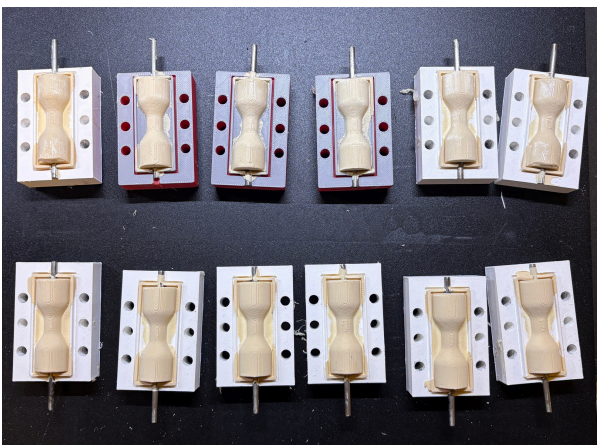
We designed the omniwheels in Autodesk Fusion 360 and printed them on a 3D printer. After printing, we assembled them using wire and silicone rings to create a functional and durable wheel system. Despite being homemade, their quality is comparable to metal omniwheels.



DRIBBLER

The dribbler consists of a MAXON motor connected by a GT2 belt with custom rubber roller.

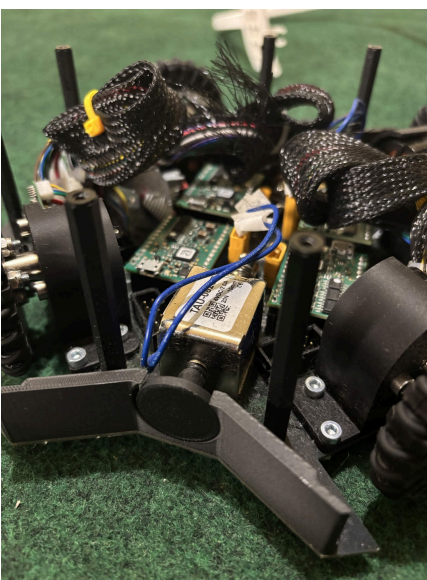
To detect whether the ball is in the dribbler, we use a mechanical contact. When the dribbler picks up the ball, it lifts up and separates the contacts.



We developed custom silicone rollers for our dribbler to improve ball control. First, we designed and 3D-printed molds in various shapes to test different surface profiles. The molds were then coated with a separating agent, and after approximately 30 minutes, filled with liquid silicone. Once the silicone cured, we tested each roller design on the robot to evaluate performance. Different shapes provided varying levels of grip and stability. One of our custom designs achieved performance comparable to silicone rollers taken from a printer mechanism. However, after testing, the original rubber rollers proved to perform better overall, so we decided to keep using them.

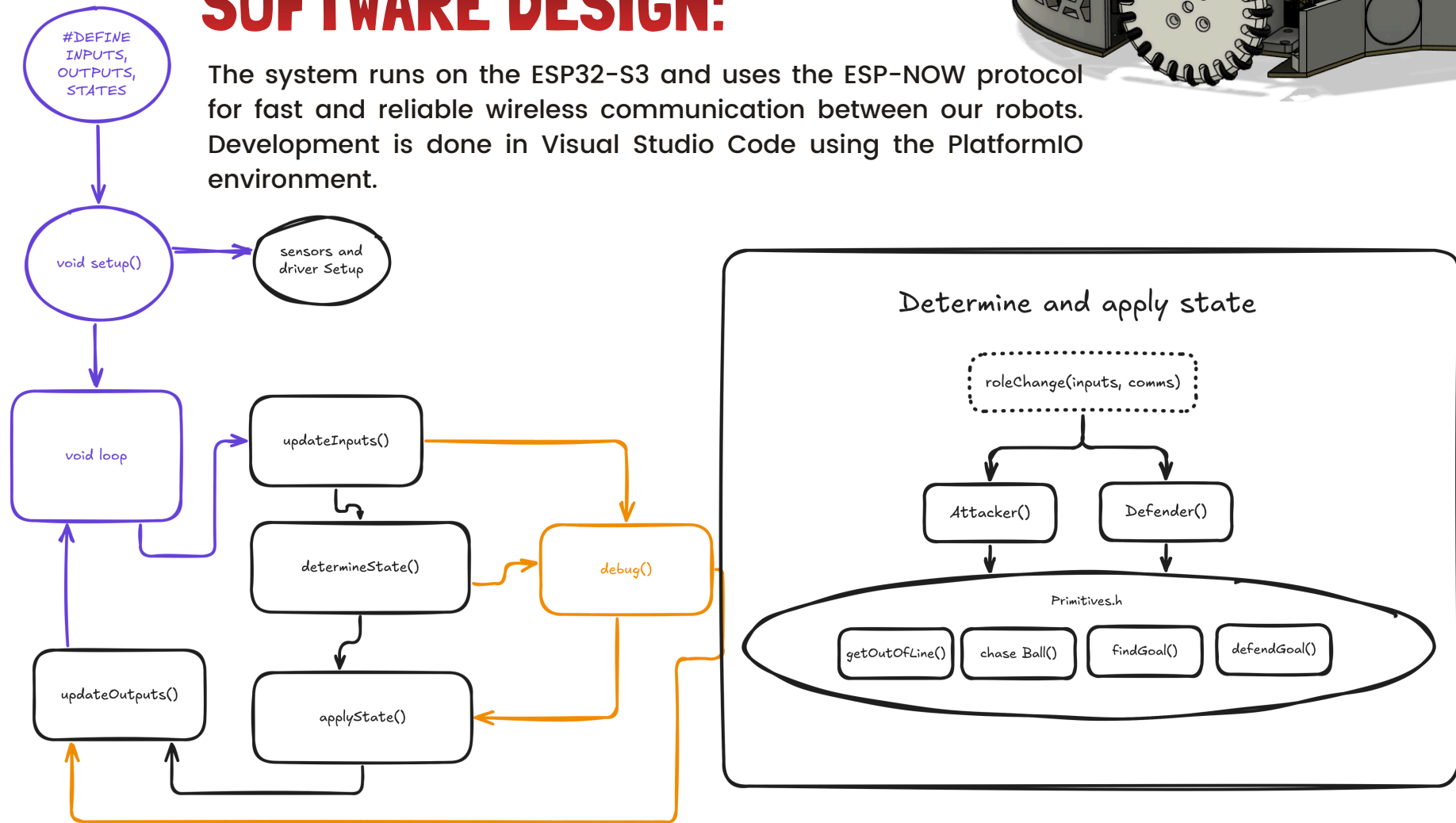
KICKER:

We use a DC-DC step-up converter to boost the voltage from 15.2V batteries to 48V, enabling a solenoid to kick a playing ball. For controlling the kicker we use mosfet with mosfet driver and additional components.



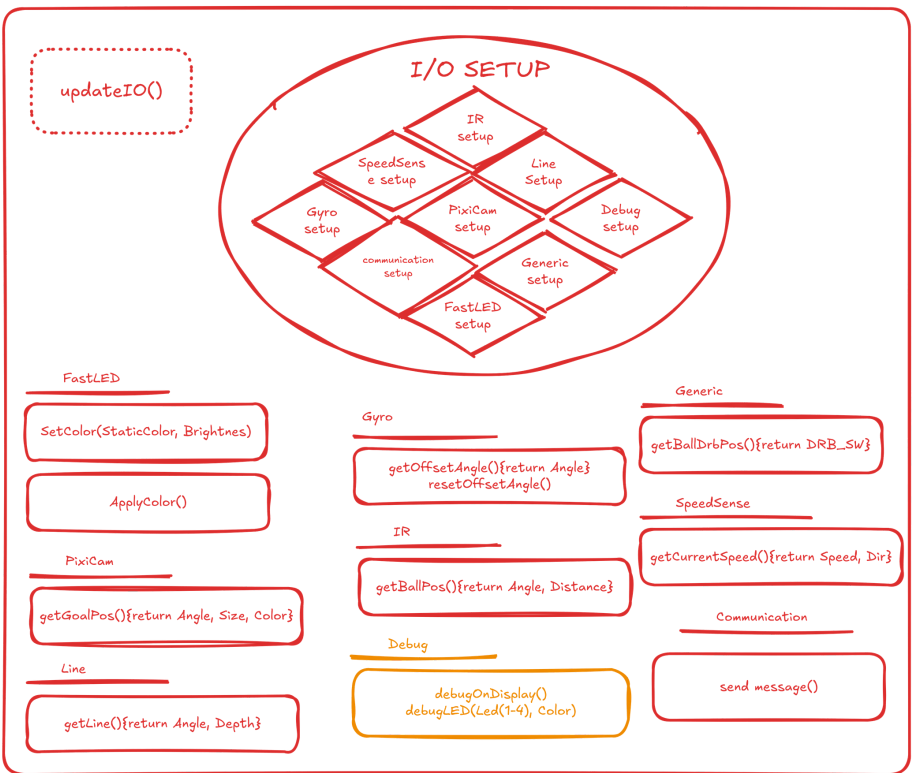
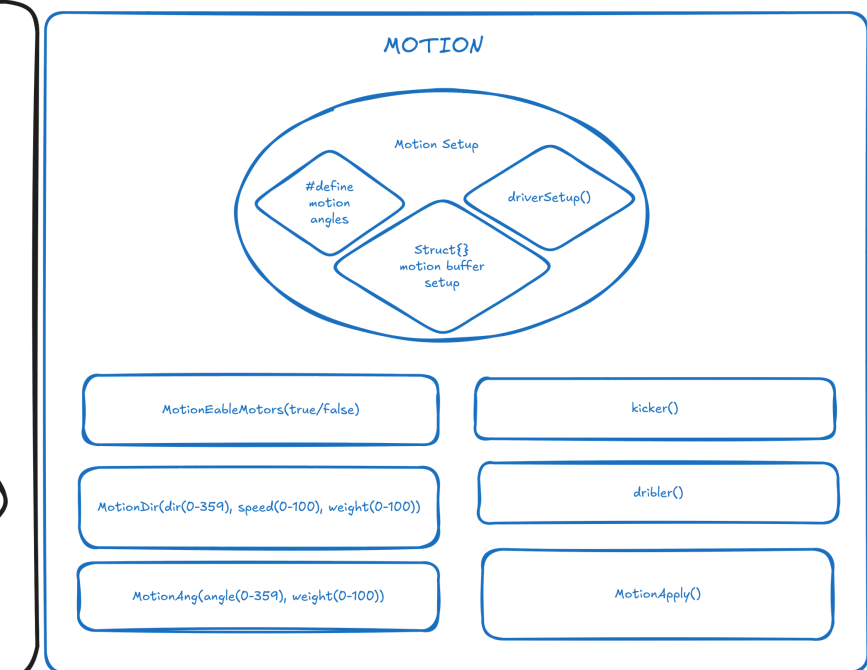
SOFTWARE DESIGN:

The system runs on the ESP32-S3 and uses the ESP-NOW protocol for fast and reliable wireless communication between our robots. Development is done in Visual Studio Code using the PlatformIO environment.



STRATEGY:

For our team strategy, we decided to assign different roles to each robot. One robot mainly works as a goalkeeper, while the second robot focuses on attacking and searching for the ball. The goalkeeper stays near our goal and defends it from opponent shots. The attacker moves across the field and tries to gain control of the ball and score. If the goalkeeper captures the ball, it can also move forward and join the attack when needed. After that, it returns to its defensive position near the goal. This strategy allows our robots to react flexibly during the match and cover both defense and offense. We believe this role distribution improves our teamwork and overall game performance.



ELECTRICAL DESIGN:

Line Sensors: To detect the field borders, we use phototransistors that sense reflected light emitted by programmable LEDs. When the robot moves over a white border line, the sensors detect a change in voltage.

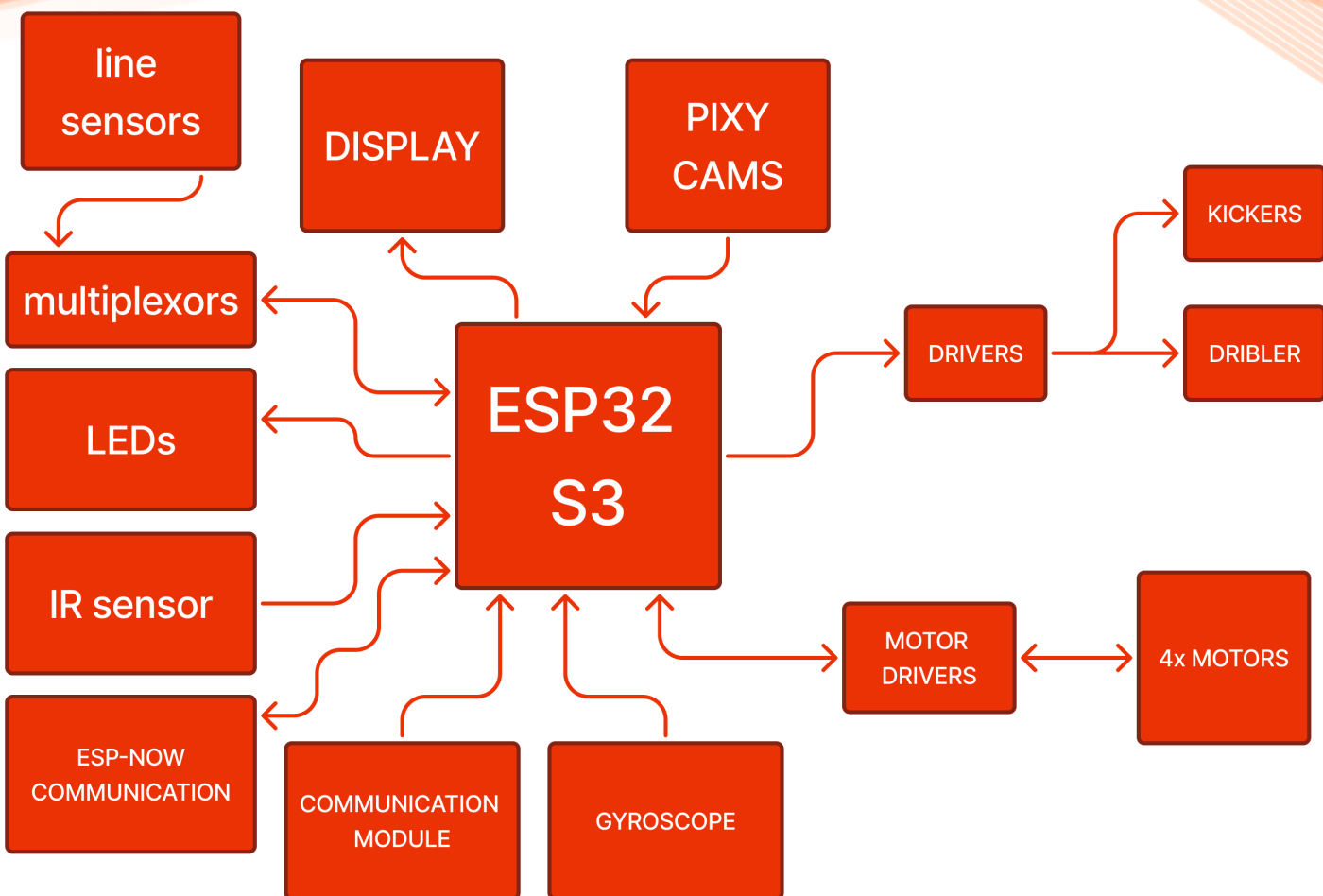
Multiplexers: Due to the limited number of analog inputs on the ESP32-S3, we use multiplexers to expand the number of line sensors we can read.

Camera: We use two Pixy Cam V2 modules for better goal detection, connected to the ESP32 via I²C.

Motor Drivers: We use ESCON 24/2 motor drivers for precise control and high power output.

Battery: We use a 4-cell lithium polymer battery, chosen for its high voltage and compact size.

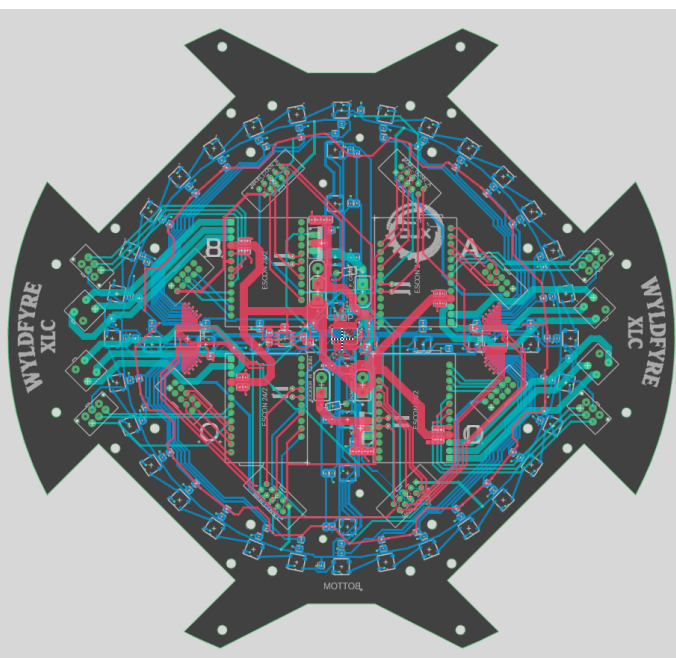
IR Sensors: To track the ball as accurately as possible, we use infrared sensors that detect the direction of the IR signal.



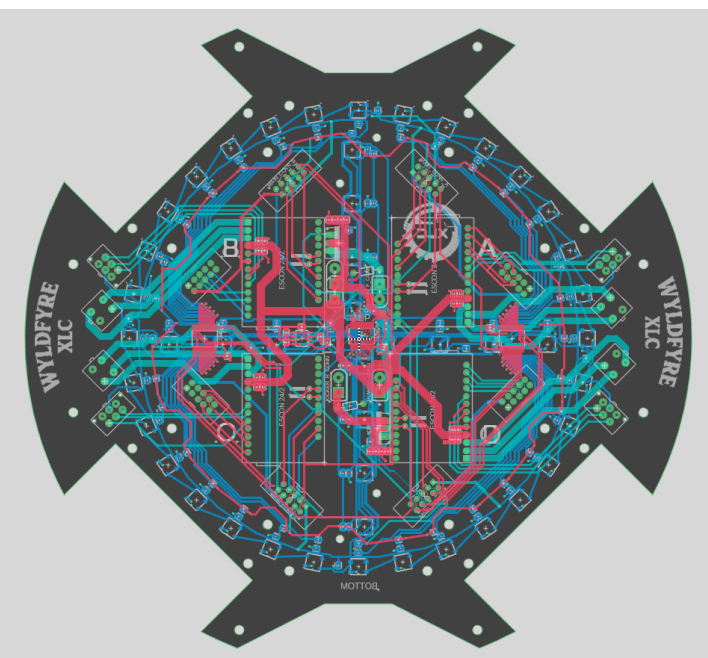
We had a challenge fitting all the components into the robot, so we designed two separate four-layer PCBs with carefully organized parts.

The design was created in Autodesk Fusion 360 to make the layout compact and efficient.

We ordered the boards from JLCPCB, which manufactured thin and lightweight PCBs according to our specifications.



BOTTOM PCB



TOP PCB

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